

1. Write a program that prints the numbers from 10 down to 1 using a for loop, similar to a countdown timer.

```
2. #include <stdio.h>
3.
4. int main() {
5.     int i;
6.
7.     for(i = 10; i >= 1; i--) {
8.         printf("%d\n", i);
9.     }
10.
11.     return 0;
12. }
```

2. Create a menu-driven console app that lets the user: 1) View your favorite 3 IPL teams, 2) Add a new team, 3) Exit. Use a while loop to keep showing the menu until the user chooses Exit.

Hint: Use input() (or Scanner in Java) to get the user's choice each time.

```
#include <stdio.h>
#include <string.h>

int main() {
    char teams[10][50] = {
        "Mumbai Indians",
        "Chennai Super Kings",
        "Royal Challengers Bengaluru"
    };

    int teamCount = 3;
    int choice;

    while (1) {
        printf("\n===== IPL Fan Menu =====\n");
        printf("1. View Favorite Teams\n");
        printf("2. Add New Team\n");
        printf("3. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);
        getchar(); // Clear newline character

        switch (choice) {
            case 1:
                printf("\nYour Favorite IPL Teams:\n");
                for (int i = 0; i < teamCount; i++) {
                    printf("%d. %s\n", i + 1, teams[i]);
                }
                break;

            case 2:
```

```

        if (teamCount < 10) {
            printf("Enter new team name: ");
            fgets(teams[teamCount], 50, stdin);

            // Remove newline character
            teams[teamCount][strcspn(teams[teamCount], "\n")] = '\0';

            teamCount++;
            printf("Team added successfully!\n");
        } else {
            printf("Team list is full!\n");
        }
        break;

    case 3:
        printf("Exiting the program...\n");
        return 0;

    default:
        printf("Invalid choice! Please try again.\n");
    }
}

return 0;
}

```

3. Build a 'Guess the Song' game like Spotify — the program randomly picks a song name from a list and asks the user to guess it. Use a do-while loop so the user can keep guessing until they get it right.

Constraint: Use at least 3 song names of your choice.

```

#include <stdio.h>
#include <string.h>
#include <stdlib.h>
#include <time.h>

int main() {
    char songs[3][30] = {
        "Believer",
        "Shape of You",
        "Perfect"
    };

    char guess[30];

    // Generate a random song
    srand(time(0));
    int randomIndex = rand() % 3;
}

```

```

printf("=== Guess the Song Game ===\n");
printf("Songs are chosen from:\n");
printf("1. Believer\n");
printf("2. Shape of You\n");
printf("3. Perfect\n\n");

do {
    printf("Guess the song: ");
    fgets(guess, sizeof(guess), stdin);

    // Remove newline character
    guess[strcspn(guess, "\n")] = '\0';

    if (strcmp(guess, songs[randomIndex]) == 0) {
        printf("🎉 Correct! You guessed the song.\n");
    } else {
        printf("❌ Wrong guess! Try again.\n");
    }

} while (strcmp(guess, songs[randomIndex]) != 0);

return 0;
}

```

4. Explain with your own example the difference between entry-controlled and exit-controlled loops by writing a short code snippet for each (for/while vs do-while) and describing what happens if the loop condition is false at the start.

Entry-Controlled Loop (for / while)

The condition is checked before the loop body executes.

If the condition is false at the beginning, the loop does not execute even once.

Exit-Controlled Loop (do-while)

The condition is checked after the loop body executes.

The loop always executes at least once, even if the condition is false initially.

While Loop:

```

#include <stdio.h>

int main() {
    int i = 5;

    while (i < 5) {
        printf("%d\n", i);
        i++;
    }

    printf("Loop finished.\n");

    return 0;
}

```

Do – While Loop

```
#include <stdio.h>

int main() {
    int i = 5;

    do {
        printf("%d\n", i);
        i++;
    } while (i < 5);

    printf("Loop finished.\n");

    return 0;
}
```